

TAC

TAC is a combination of 3 chemotherapy drugs used to treat breast cancer.

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What is TAC?

TAC is a combination of chemotherapy drugs. It is named after the initials of the drugs used for treatment. The drugs are:

- T – [docetaxel](#) (also called Taxotere[®])
- A – [doxorubicin](#) (also called Adriamycin[®])
- C – [cyclophosphamide](#).

TAC is used to treat [breast cancer](#). It is best to read this information with our general information about [chemotherapy](#) and the [type of cancer](#) you have.

Your doctor will talk to you about this treatment and its possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How TAC is given

You will be given TAC in the chemotherapy day unit. A chemotherapy nurse will give it to you.

During treatment you usually see a:

- cancer doctor
- chemotherapy nurse or a specialist nurses
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

Before or on the day of treatment, a nurse or person trained to take blood (phlebotomist) will take a blood sample from you. This is to check that it is safe for you to have chemotherapy.

You will speak to a doctor, nurse or pharmacist before you have chemotherapy. They will talk to you about your blood results and ask you how you have been feeling. If your blood results are okay, the pharmacy team will prepare your chemotherapy.

You will usually be given [steroid](#) tablets to take for 3 days, starting from the day before your chemotherapy. These help to prevent an allergic reaction to docetaxel and to reduce some other side effects. It is important to take these tablets. If you have not taken them, tell your doctor, nurse or pharmacist before you start your chemotherapy. Some people may have the steroid as a drip on the day of treatment instead of tablets.

Your nurse usually gives you anti-sickness (anti-emetic) drugs before the chemotherapy.

You may have the chemotherapy drugs through:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in your arm or hand
- a [central line](#) – a fine tube that goes under the skin of your chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in your arm and goes up into a vein in your chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on your chest or arm and goes into a vein in your chest.

Your nurse will give you doxorubicin (a red fluid) and cyclophosphamide as slow injections. These are given directly into your vein, with a drip (infusion) to flush them through. Each takes around 5 to 10 minutes.

Sometimes, cyclophosphamide is given as a drip over 30 minutes into your cannula or line. Docetaxel is also given as a drip, which usually takes about an hour.

Your nurse or doctor will talk to you about how the treatment will be given.

Your course of chemotherapy

You usually have a course of several cycles of treatment over a few months.

Each cycle of TAC takes 21 days (3 weeks). On the first day, you have all 3 chemotherapy drugs. Then you will have no chemotherapy for the next 20 days.

At the end of the 21 days, you will start your second cycle of TAC. This is the same as the first cycle. Your doctor or nurse will tell you how many cycles you are likely to have.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Side effects while treatment is being given

Some people may have side effects while they are having a cancer drug or shortly after they have it:

Allergic reaction

Some people have an allergic reaction while having this treatment. Before treatment, you will have medicines to help prevent or reduce any reaction.

Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick

- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

The drug leaks outside the vein

Sometimes cancer drugs that are given into a vein may leak outside the vein. If this happens, some drugs can damage the tissue near the vein. This is called extravasation. Extravasation is not common, but it is important that it is dealt with quickly. If you have any of the following symptoms around the vein during or after your treatment, tell your nurse straight away:

- stinging
- pain
- swelling
- if you have white skin, the area may become red
- if you have black or brown skin, the area may become darker.

Pain along the vein

This treatment can cause pain:

- at the place where the drip (infusion) is given
- along the vein.

If you feel pain, tell your nurse straight away. They can check the site. They may give the drug more slowly or flush it through with more fluid to reduce pain.

Flushes and blocked nose

Some people who have cyclophosphamide have hot flushes or feel like their nose is blocked while the drug is being given. You may also have a strange taste in your mouth. These side effects do not last for long. But if you notice this, ask the nurse to slow down the drip. This will reduce these symptoms.

Facial flushing

You may suddenly feel warm and your face may get red when you have doxorubicin. This should only last a few minutes.

Common side effects

Risk of infection

This treatment can reduce the number of white blood cells in your blood. These cells fight infection. If your white blood cell count is low, you may be more likely to get an [infection](#). A low white blood cell count is called neutropenia.

An infection can be very serious when the number of white blood cells is low. It is important to get any infection treated as soon as possible. If you have any of the following symptoms, contact the hospital straight away on the 24-hour number:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature
- you have symptoms of an infection.

Symptoms of an infection include:

- feeling shivery and shaking
- a sore throat
- a cough
- breathlessness
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

It is important to follow any specific advice your cancer treatment team gives you.

Your white blood cell count will usually return to normal before your next treatment. You will have a blood test before having more treatment. If your white blood cell count is low, your doctor may delay your treatment for a short time, until your cell count increases.

You may be given a drug called [G-CSF](#). This encourages the body to make more white blood cells. You have it as a small injection under the skin.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
-

tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Bladder problem

Cyclophosphamide may irritate your bladder and cause discomfort when you pass urine (pee). The advice is usually to try to drink at least 2 litres (3½ pints) of fluid during the 24 hours after chemotherapy. But follow any advice from your doctor, nurse or pharmacist about how much is right for you.

It is also important to empty your bladder regularly and to try to pass urine as soon as you feel the need to go.

Contact the hospital on the 24-hour number straightaway if you have:

- discomfort or stinging when you pass urine
- any blood in your urine.

Red or pink urine (pee)

After doxorubicin, your urine may be pink or red for up to 48 hours. This is because of the red colour of the drug. It is not harmful. But always check with your doctor, nurse or pharmacist if you are worried.

Build-up of fluid

You may put on weight or your ankles, legs and hands may swell. This is because of fluid building up. The [steroid](#) tablets your doctor prescribes will help to prevent this. Tell your doctor or nurse if fluid builds up. If your ankles and legs swell, it can help to put your legs up on a foot stool or cushion. The swelling gets better after your treatment ends.

Feeling sick

Your doctor, nurse or pharmacist will prescribe anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to, even if you do not feel sick. It is easier to prevent sickness than to treat it after it has started.

If you feel sick, take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids. If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice. They may change your anti-sickness treatment. Let them know if you still feel sick.

Diarrhoea

This treatment may cause severe diarrhoea. Diarrhoea means passing more stools (poo) than is normal for you, or having watery or loose stools. If you have a stoma, it will be more active than usual.

Your doctor, nurse or pharmacist may give you anti-diarrhoea drugs to take at home.

If you have diarrhoea or a mild increase in stoma activity, follow their advice about:

- taking anti-diarrhoea drugs
- how much and what type of fluids to drink
- any changes to your diet that might help.

Contact the hospital straight away on the 24-hour number if any of the following things happen:

- you have diarrhoea at night
- you have uncomfortable stomach cramps
- you have diarrhoea 4 or more times in a day
- you have a moderate increase in stoma activity
- the anti-diarrhoea drugs do not work within 24 hours.

You may need to give a stool sample, or go to hospital to have fluids through a drip or antibiotics.

Sore mouth and throat

This treatment may cause a sore mouth and throat. You may also get mouth ulcers. This can make you more likely to get a mouth or throat infection. Use a soft toothbrush to clean your teeth or dentures in the morning, at night and after meals.

Contact the hospital straight away on the 24-hour number, if:

- a sore mouth or throat affects how much you can drink or eat
- your mouth, tongue, throat or lips have any blisters, ulcers or white patches.

They can give you advice, and mouthwash or medicines to help with the pain or to treat any infection. Follow their advice and make sure you:

- drink plenty of fluids
- avoid alcohol and tobacco
- avoid food or drinks that irritate your mouth and throat.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Feeling tired

[Feeling tired](#) is a common side effect of this treatment. It is often worse towards the end of treatment and for some weeks after it ends. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

If you feel sleepy, do not drive or use machinery.

Hair loss

You usually lose all the hair from your head. You may also lose your eyelashes and eyebrows, as well as other body hair. [Hair loss](#) usually starts after your first or second treatment.

[Scalp cooling](#) is a way of lowering the temperature of your scalp to help reduce hair loss. Your nurse can tell you whether this is an option for you.

If you want to cover up hair loss, there are different ways to do this. Your nurse can give you information about coping with hair loss. Remember to [protect your skin from the sun](#). Use sunscreen with a sun protection factor (SPF) of at least 30 on your scalp. Or cover up with a hat or scarf.

Hair loss is usually temporary. Your hair will usually grow back after treatment ends. Very rarely, hair may not grow back. Or it may grow back thinner than before. If you are worried about this, talk to your doctor, nurse or pharmacist.

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Muscle or joint pain

You may get pain in your muscles or joints for a few days after treatment. If this happens, tell your doctor, nurse or pharmacist. They can give you painkillers and advice. They can also tell you if any of the painkillers you usually take are suitable.

If you have muscle or joint pain, try:

- placing a heat pad or covered hot water bottle against the painful area
- taking warm baths
- planning your activities to include regular rests.

Skin changes

TAC may affect your [skin](#). Your doctor or nurse can tell you what to expect. If your skin feels dry, try using an unperfumed moisturising cream every day. During treatment and for several months afterwards, you will be more sensitive to the sun and your skin may burn more easily than usual. You can still go out in the sun, but remember to:

- use a sun cream with a sun protection factor (SPF) of at least 30
- cover up with clothing and a hat.

Your skin may darken. It will return to its normal colour after you finish treatment. If you have had [radiotherapy](#), either recently or in the past, the area that was treated may become red or sore.

Rarely, a more serious skin condition can develop. You may have a skin rash which then blisters, and your skin can peel. You may also feel unwell with flu-like symptoms, such as a high temperature and joint pain. If you have any of these symptoms, contact your doctor or hospital straight away.

Always tell your doctor or nurse about any skin changes. They can give you advice and may prescribe creams or medicines to help. Any changes to your skin are usually temporary and improve when treatment finishes.

Hand-foot (palmar-plantar) syndrome

This treatment can affect the palms of your hands and the soles of your feet. This is called palmar-plantar or hand-foot syndrome.

If you have white skin these areas may become red. If you have black or brown skin, these areas might get darker.

The skin on the palms of your hands and the soles of your feet may:

- be sore
- be painful, tingle, or swell
- peel, crack or blister.

If you have any of these symptoms, contact the hospital straight away on the 24-hour number. They can give you advice. This is especially important if you have any broken skin or if walking is difficult. They can prescribe creams and painkillers to help.

You can care for your hands and feet by:

-

keeping your hands and feet cool by washing in cool water

- gently moisturising your hands and feet regularly
- wearing gloves to protect your hands and nails when working in the house or garden
- wearing loose cotton socks and avoiding tight-fitting shoes and gloves.

Nail changes

This treatment can affect [your nails](#). They may grow more slowly or break more easily. You might notice ridges or white or dark lines across your nails. These changes usually disappear as the nails grow out after treatment. Sometimes nails can become loose or fall off. Tell your doctor, nurse or pharmacist about any changes to your nails.

If the skin around your nails becomes sore and swollen, contact the hospital straight away on the 24-hour number. These might be signs of an infection.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Eyesight changes

This treatment can affect your eyes and eyesight. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- dry, itchy, painful, or swollen eyes
- eyes that are sensitive to light
- changes to your eyesight.

Effects on the heart

This treatment can affect how the heart works. You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

If the treatment is causing [heart problems](#), your doctor may change the type of treatment you are having.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness

- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the lungs

This treatment can cause changes to the lungs. Tell your doctor, nurse or pharmacist if you develop:

- a cough that does not go away
- wheezing
- breathlessness.

You should also tell them if any existing breathing problems get worse. You may have tests to check your lungs.

Less common side effects

Effects on the kidneys and liver

This treatment can affect how your kidneys and liver work. This is usually mild and goes back to normal after treatment. You will have blood tests to check how well your kidneys and liver are working.

It is important to drink at least 2 litres (3½ pints) of fluids each day to help protect your kidneys.

Second cancer

This treatment can increase the risk of developing a second cancer years later. This is rare. But the benefits of treatment usually far outweigh this risk. Your doctor can talk to you about this.

Other information

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
-

reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker

- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Food and medicines

During treatment with cyclophosphamide you should avoid grapefruit and grapefruit juice. You should also not use the herbal remedy St John's Wort. This is because they can make cyclophosphamide less effective and may increase side effects.

Some medicines can affect how this treatment works or be harmful when you are having it. Always tell your cancer doctor about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop or chemist
- vitamins, herbal drugs and complementary therapies.

Tell other doctors, pharmacists or dentists who prescribe or give you medicines that you are having this cancer treatment.

You can visit the [electronic Medicines Compendium \(eMC\)](#) for more detailed information about your treatment.

Alcohol

Some preparations of this treatment contain alcohol. If having alcohol is a problem for you, tell your doctor, nurse or pharmacist. Your blood alcohol level may be above the legal limit after you have the treatment. Do not drive or operate machinery for a few hours after having this treatment, even if you feel okay.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Changes to periods

If you have periods, these may become irregular or stop while you are having this treatment. They might return after treatment, but this does not always happen. Your [menopause](#) may start sooner than it would have done. Your doctor, nurse or pharmacist can give you more information.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Sex

It is possible that small amounts of chemotherapy may be passed on through vaginal fluids or semen. If you have sex in the first few days after treatment, your cancer team will usually advise using condoms or a dental dam to protect your partner.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

About our information

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert health professionals and people living with cancer.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 November 2023 | Next review: 01 November 2025

⚠ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information.](#)

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help